Zoo-515P BIOLOGICAL TECHNIQUE

- EFFORTS MADE BY:
- AYESHA BUTT (BS ZOOLOGY)
- It covers the PPTs of Practical number# 4,5 6,7,8,9,11,

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Practical # 04

Measurement Cell Size: Prokaryotic (Bacterial Cell)

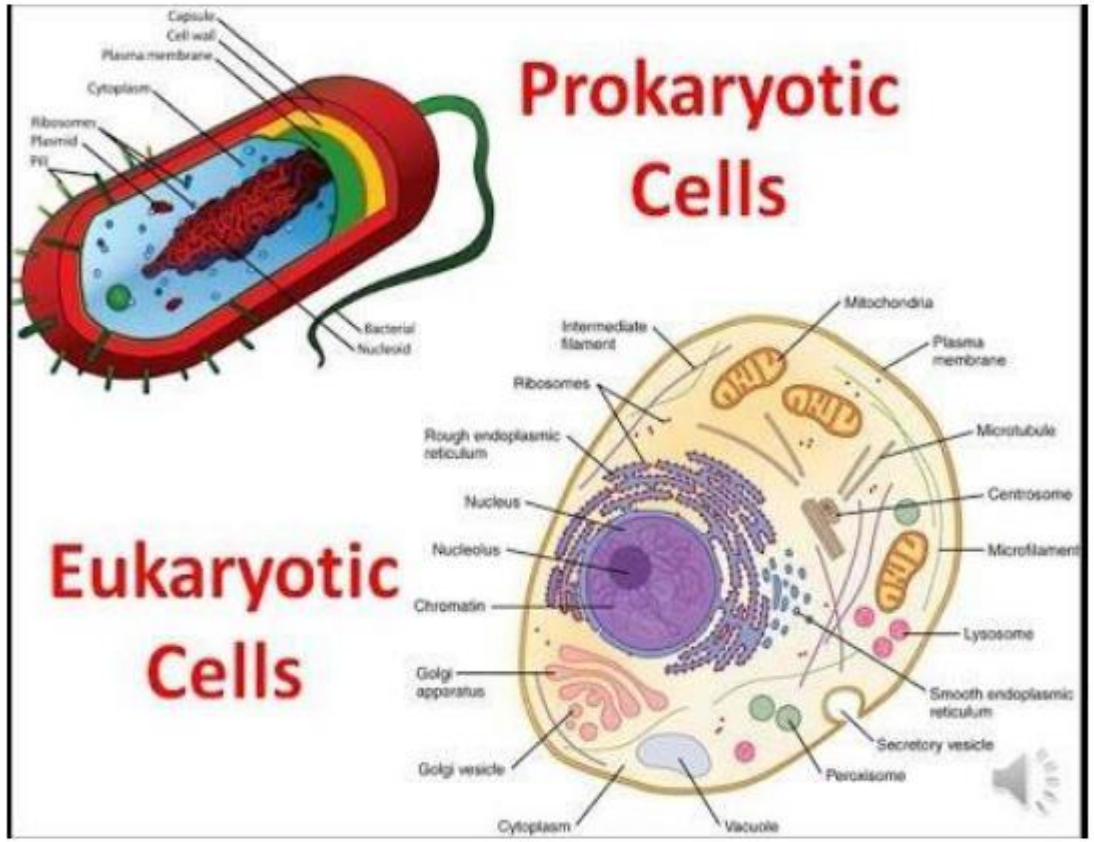
Learning objectives

- To learn the students about preparing the slides of human cheek cells
- To learn the students to determine the field of view of microscope
- To learn the students to estimate the size of the cell

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What are eukaryotes?

- A membrane-bound nucleus.
- Generally larger and more complex than prokaryotic cells.
- For example, of protists, fungi, plants, and animals are all eukaryotic in nature.
- Typical size: 10 to 100 µm in diameter



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Materials and Reagents required

- Microscope
- Glass slides
- Glass cover slips
- Methylene blue
- Toothpicks
- Distilled water
- Discarding box
- White board and board marker
- Labelling marker
- Pipette and pipette tips
- Gloves
- Methylated spirit

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Steps of cell measurement

Step: 01

Preparation of slides of human cheek cells

Step: 02

Estimating the Field of view/ field diameter for microscope

Step: 03

Observing the specimen and size under microscope

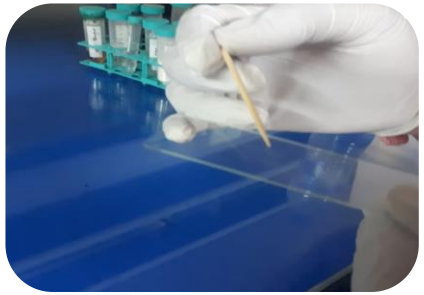
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1. Preparing slide of human cheek cells

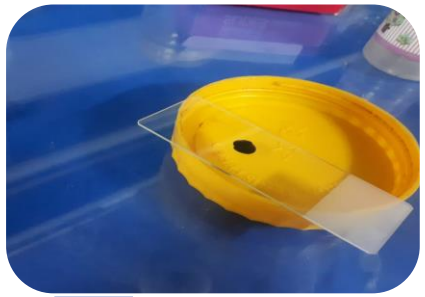
i. Place a drop of distilled water on glass slide
ii. Gently scrap the internal lining of cheek with toothpick carefully
iii. Mix in water drop already placed on glass slide and make a smear
iv. Add drop of methylene blue
v. Observe under microscope



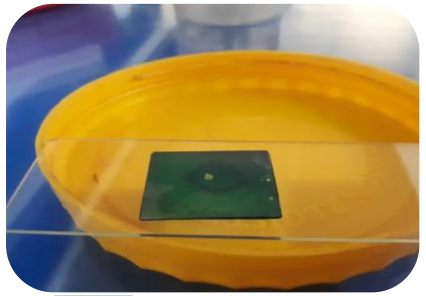
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- Preparation of smear



- Staining with methylene blue



- Applying the cover slip



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2. Estimating the Field of view/ field diameter for microscope

- Field of view= Field number/Objective magnification



Field of view = 20/10

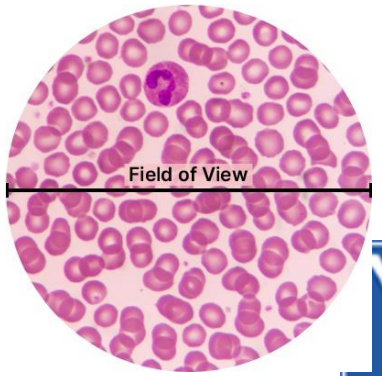
Field of view = 2 mm

1mm=1000 µm

Therefore,

2 mm or 2000 µm

Higher the magnification, lower the field diameter



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3. Observing the specimen and size under microscope

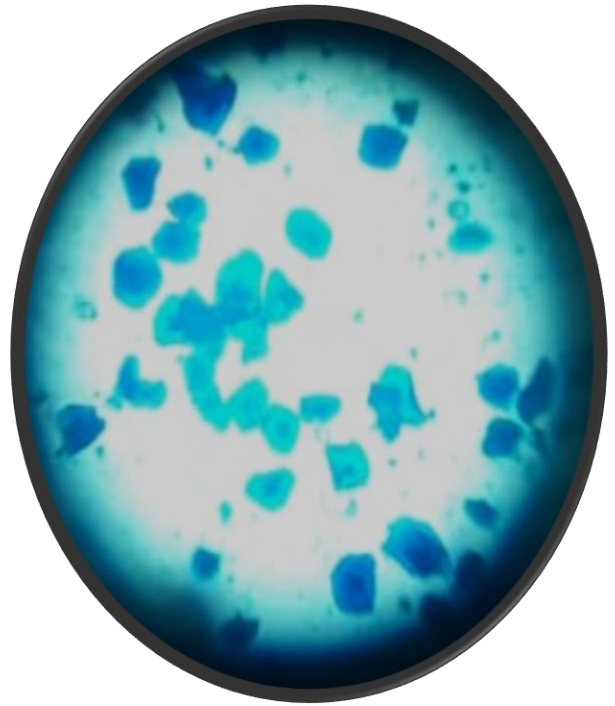
Size of specimen= Diameter of field view/ Number of cells cross the diameter

For example,

Diameter of field of view= 2000 µm
Number of cells= 05

Size= 2000/05

Size of the specimen or cell= 400 µm



2000 µm

Practical # 05

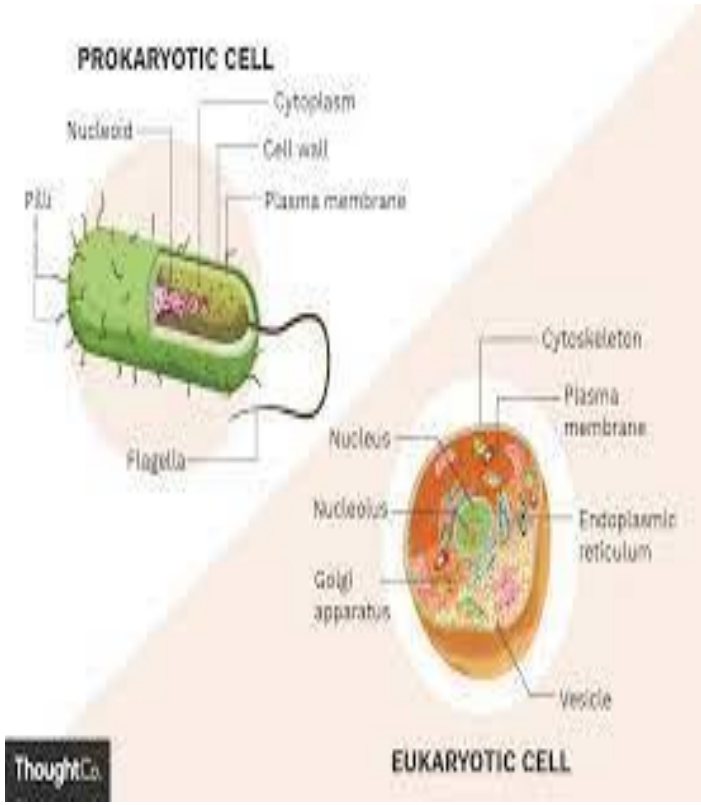
Measurement of cell Size: Eukaryotic cell (Human cheek cells)

Introduction

- A cell is the smallest unit of a living thing. A living thing, whether made of one cell (like bacteria) or many cells (like a human), is called an organism. Thus, cells are the basic building blocks of all organisms. Cells fall into one of two broad categories:

- **Prokaryotic:** Only the predominantly single-celled organisms of the domains Bacteria and Archaea are classified as prokaryotes (pro- = “before”; -kary- = “nucleus”)

- **Eukaryotic:** Cells of animals, plants, fungi, and protists are all eukaryotes (eu- = “true”) and have a nucleus.



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Components of prokaryotic cells:

- All cells share four key components:

1. Plasma membrane:

- It is an outer covering that separates the cell‘s interior from its surrounding environment

2. Cytoplasm

- It consists of the jelly-like cytosol inside the cell, plus the cellular structures suspended in it. In eukaryotes, cytoplasm specifically means the region outside the nucleus but inside the plasma membrane.

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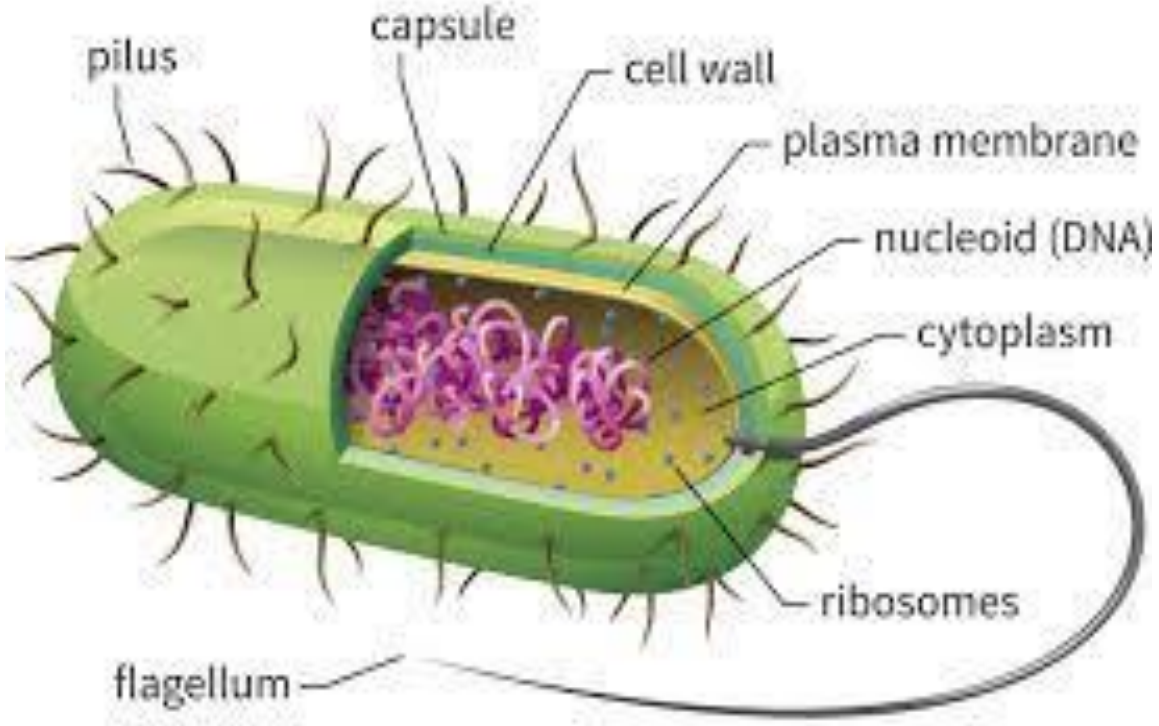
3. DNA

- It is the genetic material of the cell

4. Ribosomes

- These are molecular machines that synthesize proteins.

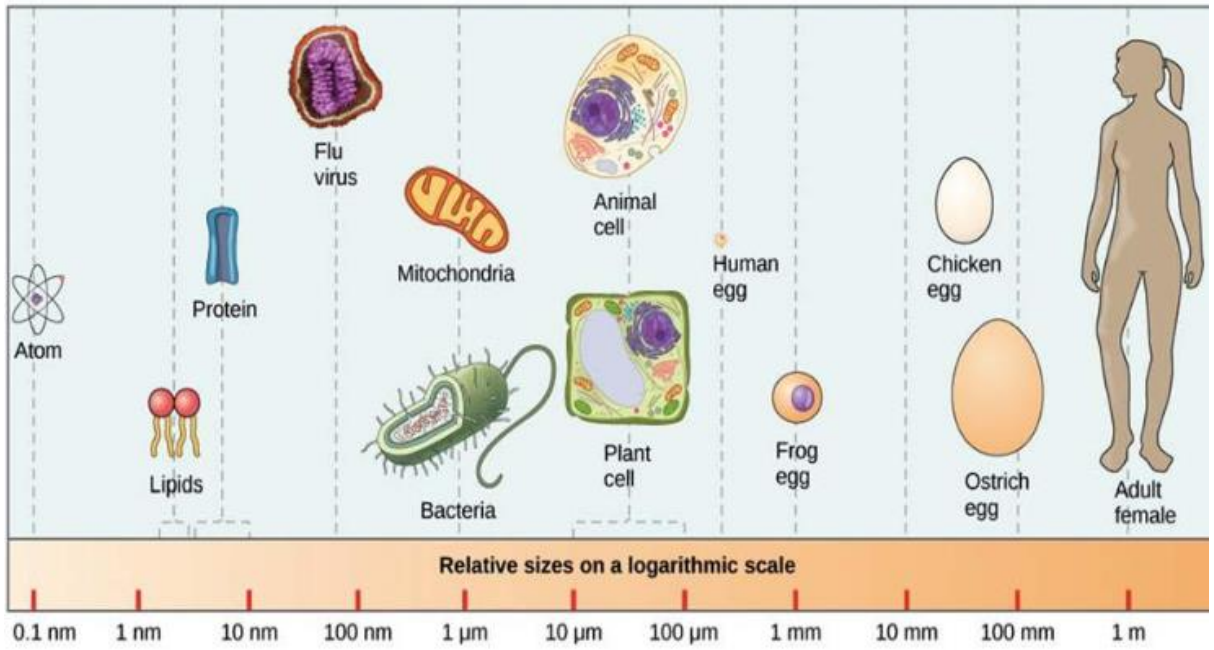
- Despite these similarities, prokaryotes and eukaryotes differ in a number of important ways. A prokaryote is a simple, single-celled organism that lacks a nucleus and membrane-bound organelles.



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Cell size

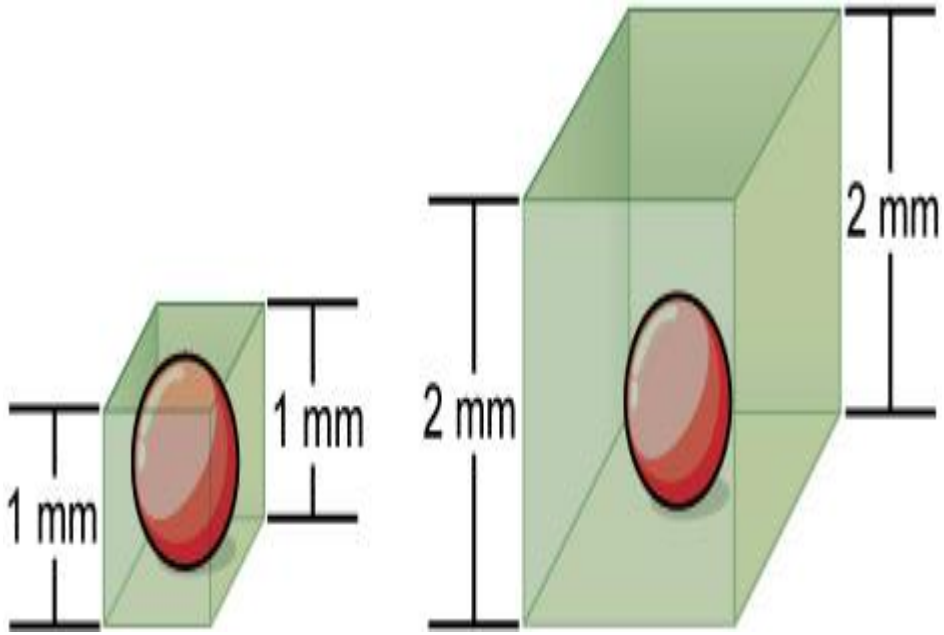
- Typical prokaryotic cells range from 0.1 to 5.0 micrometers (μm) in diameter
- significantly smaller than eukaryotic cells, which usually have diameters ranging from 10 to 100μm.



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Estimating cell size

- Small size, in general, is necessary for all cells, whether prokaryotic or eukaryotic.
- Let’s examine why that is so.
- First, we’ll consider the area and volume of a typical cell. Not all cells are spherical in shape, but most tend to approximate a sphere.
- You may remember from your high school geometry course that the formula for the surface area of a sphere is $4\pi r2$, while the formula for its volume is $4\pi r2/3$.
- Thus, as the radius of a cell increases, its surface area increases as the square of its radius, but its volume increases as the cube of its radius (much more rapidly).
- Therefore, as a cell increases in size, its surface area-to-volume ratio decreases. This same principle would apply if the cell had the shape of a cube.
- If the cell grows too large, the plasma membrane will not have sufficient surface area to support the rate of diffusion required for the increased volume.
- In other words, as a cell grows, it becomes less efficient. One way to become more efficient is to divide; another way is to develop organelles that perform specific tasks.



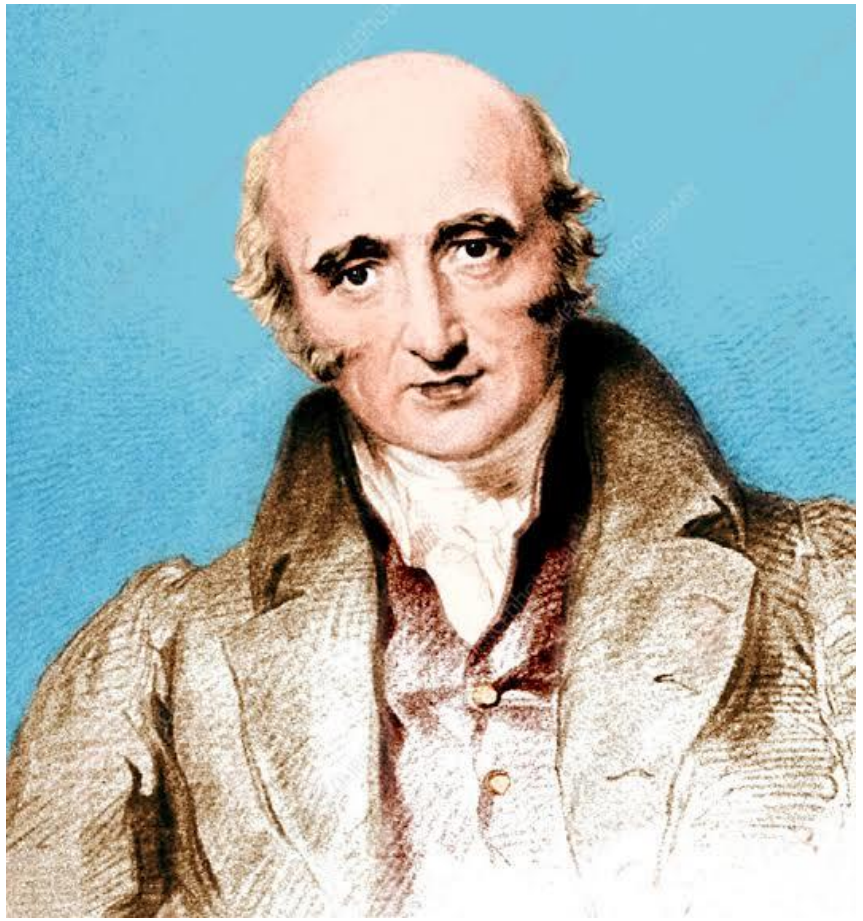
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Practical # 06

Recording of Microscopic Observations with the Help of Camera Lucid

What is camera lucida?

- Camera Lucida, when attached with a compound microscope, helps drawing microscope images of objects on paper.
- Works on simple optical principle reflecting beam of light through a prism and a plane mirror
- Invented in 1806 by William Hyde Wollaston and Christian Gobrecht



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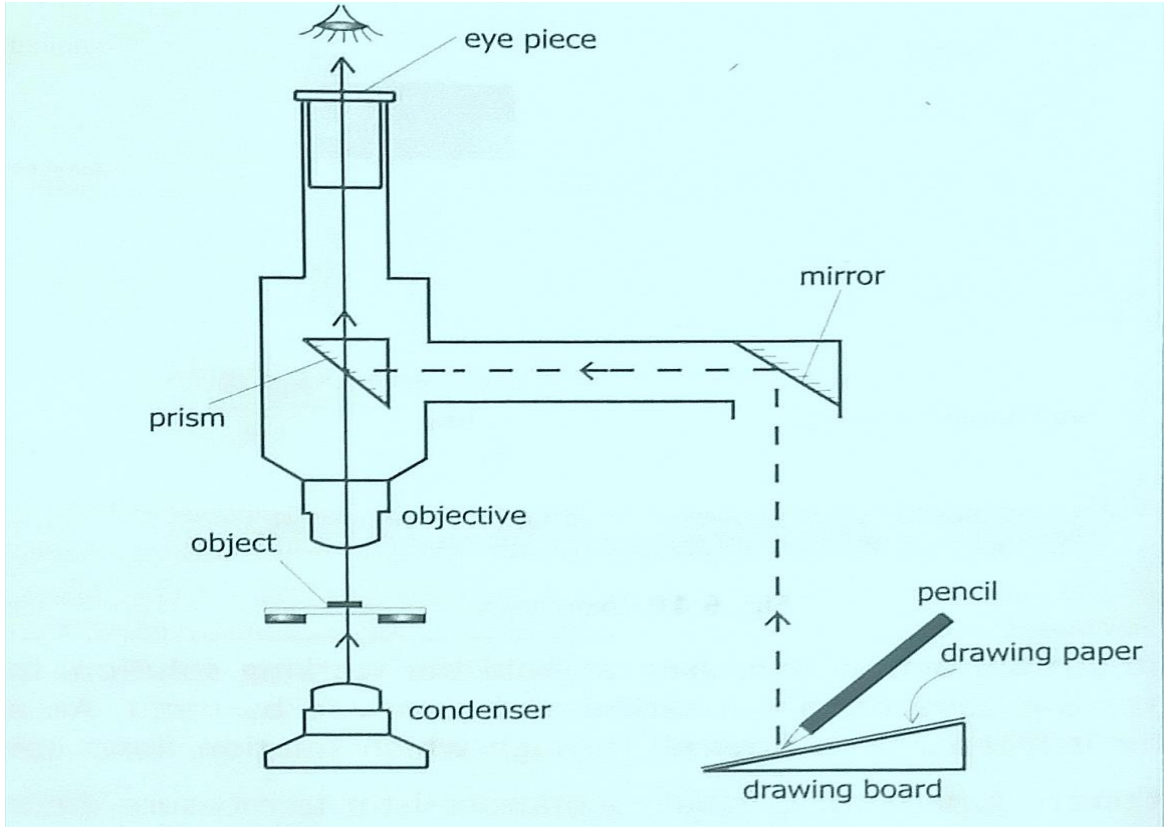


Camera lucida attached to microscope

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Parts of camera lucida

1. The attachment ring
2. The prism
3. The mirror

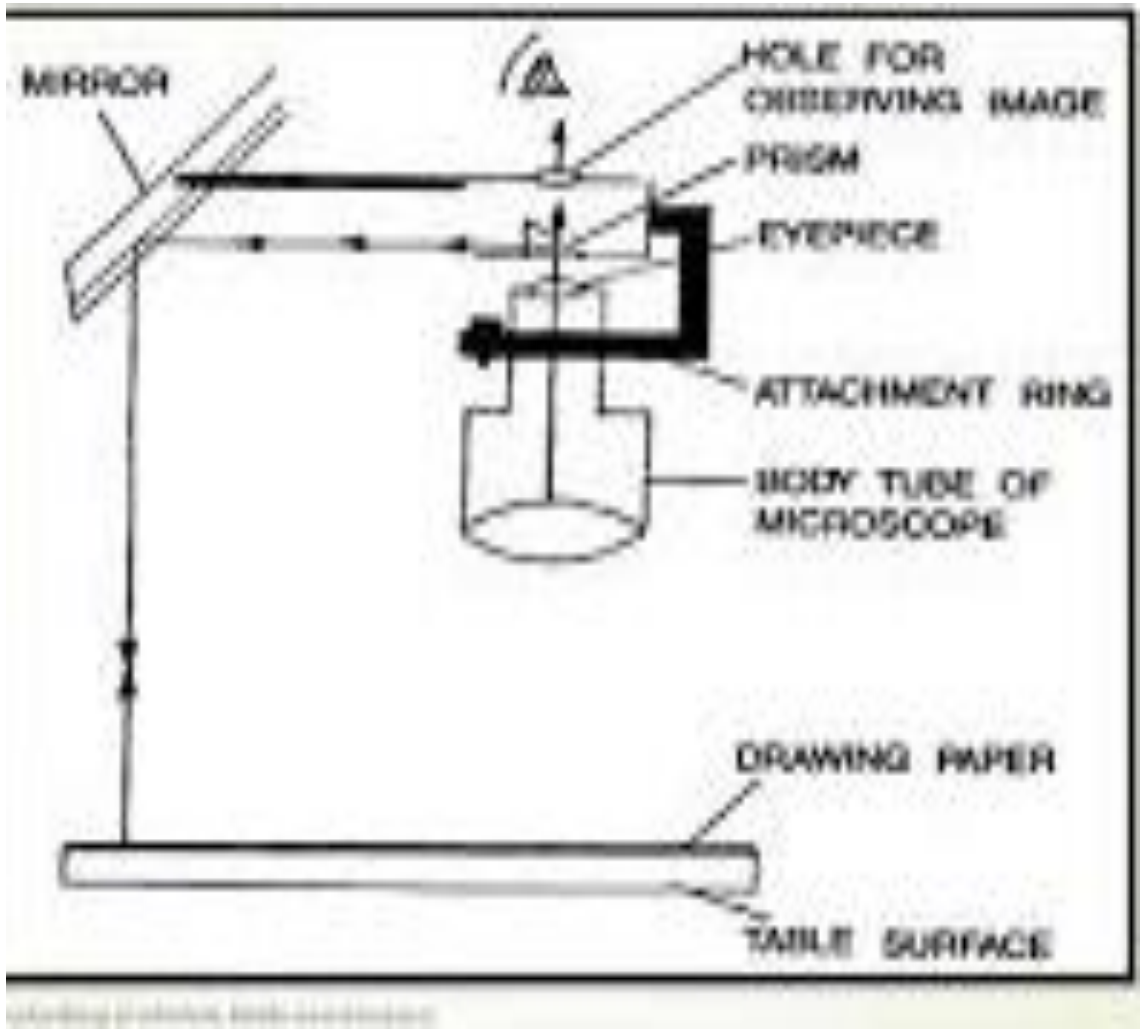


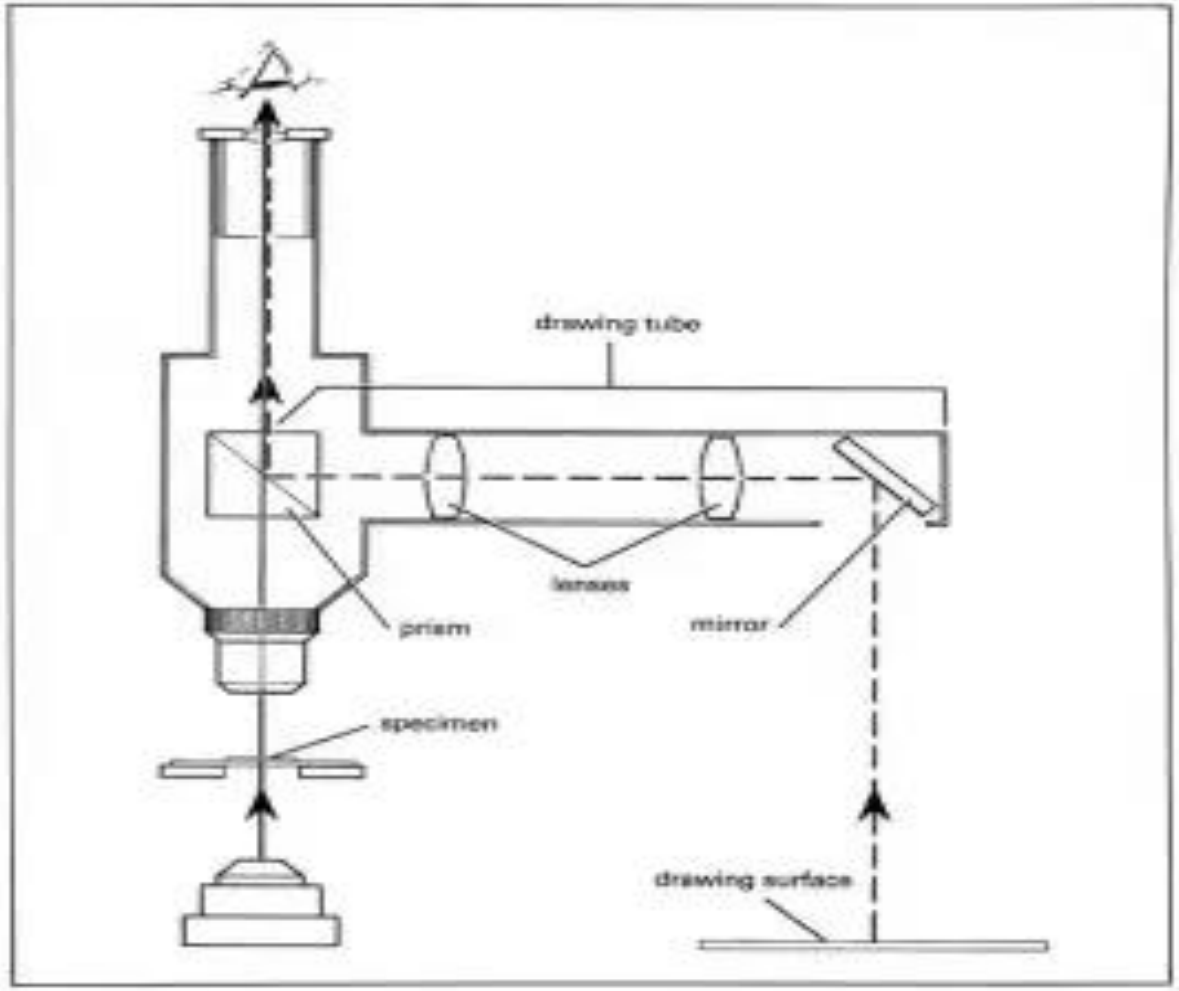
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How it works?

- The attachment ring attaches the lucida of the camera and the body tunnel to the microscope.
- The prism sits above the eyeball when the tool is attached to a microscope.
- The observer now sees a picture of an object under a microscope using a prism that displays the image horizontally on the plane screen.
- The plane's mirror, attached to the end of the arm, rotates, and is positioned at a 45 ° angle to the prism and plane paper.

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Schematic diagrams of camera lucida

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Practical # 07

Liquid Handling: Proper Use of Pipettes

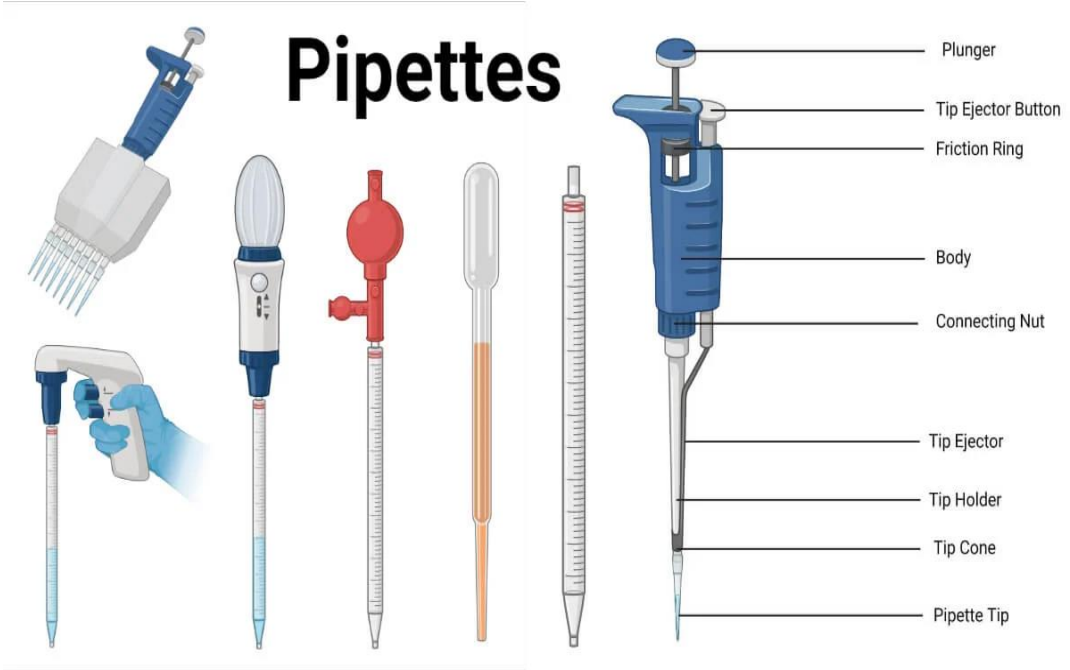
What is pipette?

- A pipette is a lab device used to measure out or dispense small amounts of liquid in volumes of milliliters (mL) or microliters (µL)
- The name “pipette” is derived from the French word “pipette,” which means “small pipe”.
- Pipettes come in three basic varieties:

i. Glass,

ii. Plastic

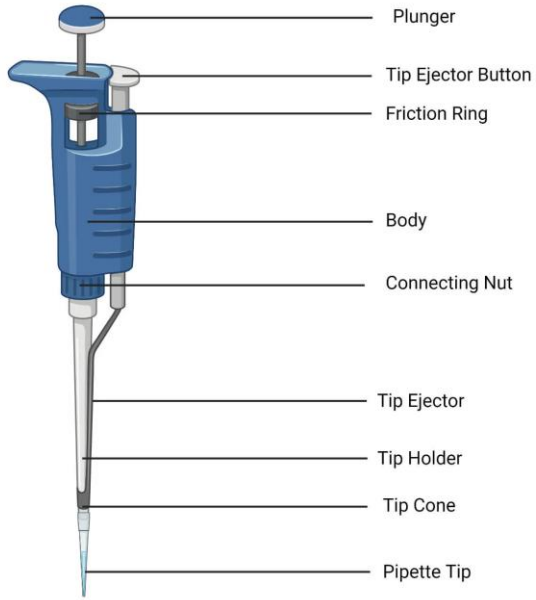
iii. Electronic.



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Parts of pipette

- **Plunger:** It is used to aspirate and dispense the desired volume of liquids.
- **Ejector button:** It causes the metal bar to descend, ejecting the pipette tip.
- **Volume Adjustment Knob:** The volume of the sample to be transferred can be set by rotating the volume adjustment knob to increase and decrease the concentration. It cannot be rotated in a fixed-volume pipette.
- **Volume window:** It displays the set volume, and the place values shown depend on pipette size.
- **Shaft:** It is a tube-like structure filled with air in an air displacement pipette. When the piston is pressed, some air from the shaft is released, aiding in liquid aspiration. In contrast, releasing the piston aids in liquid dispensing while refilling the shaft with air.
- **Tip cone:** It is the area where the pipette tip is attached.
- **Pipette tip:** It is a piece of equipment made of polypropylene or plastic that has been molded into it and comes into contact with the liquid.



Parts of a Pipette

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Types of pipettes

- Disposable/Transfer Pipette
- Graduated/Serological Pipette
- Single-Channel Pipette
- Multichannel Pipette
- Repeat Pipette Dispenser

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Disposable/Transfer Pipette

- Most basic type of pipette
- Used for rough measurement
- Aspirate liquid at a 90-degree angle, dispense at a 45-degree angle, and touch off to make sure all the liquid is dispensed.



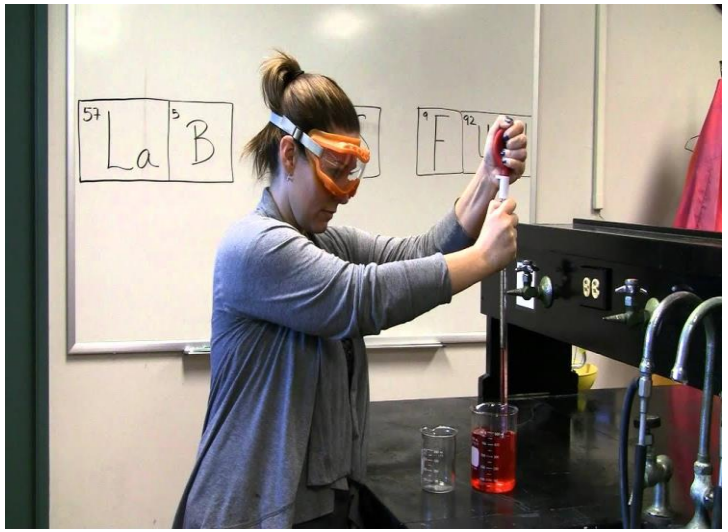


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Graduated/Serological Pipette

- Hold pipette in solution, don’t touch the bottom.
- Squeeze the bulb and attach to the top of the pipette.
- Hold forefinger on top of pipette to control volume aspiration.
- Subtract the amount needed into a separate beaker while staying eye level to assure proper measurement.
- Measure solution from the bottom of the meniscuses, the crescent-shaped surface of the liquid that is visible in the pipette.
- Subtract needed volume from the initial volume and find the volume needed to release to in order to get the desired amount.





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Single-Channel Pipette



- an air-displacement design which produces accurate measurement results with the use of one disposable tip
- Two techniques:

Forward Technique

- The most common technique for pipette measurements.
- Press the plunger to first stop and slightly submerge the pipette tip into the liquid,
- Aspirate your measured volume by releasing the plunger slowly to prevent bubbles.
- To dispense the liquid, place the tip against the side of the receptacle, then slowly press the plunger through the first stop to the final blow out position while 'touching off' the last drop from the tip.

Reverse Technique

- When working with viscous solutions or solutions prone to bubbles we have the option to use a reverse pipetting technique, which will minimize interference from air bubbles.
- To use this technique, press plunger all the way through to the third stop position (all the way down), slightly submerge into liquid, and slowly release the plunger all the way back to the top and aspirate the liquid into the tip.

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Multichannel Pipette

- a single channel, except it takes more than one tip at a time
- Install of the proper tips to each channel and set the desired volume.
- Hold the pipette in a vertical position, depress the plunger to the first stop.
- Immerse the tip into the liquid, release the plunger back to the rest position.
- Place the tip 45 degrees against the wall of the vessel receiving the liquid.
- Depress the plunger to the first stop, wait one second, press the plunger to the second stop and expel all the liquid while ‘touching off’ the last drop.
- Move the end of the tip away from the liquid and release the plunger to the rest position.



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Repeat Pipette Dispenser

- This type of pipette allows a technician to set and dispense a specific volume into multiple receptacles without having to aspirate in between dispenses
- Slide the filling lever down as far as it goes.
- Raise the locking clamp upward.
- Insert the syringe-type tip into the barrel so that it clicks into position, then close the lever.
- Immerse the tip into the liquid at a 90 degree angle.
- Slowly slide the filling lever upward to fill the tip completely.
- Prime the tip by discarding the liquid from the first dispense.
- The repeat pipette is now ready to operate.



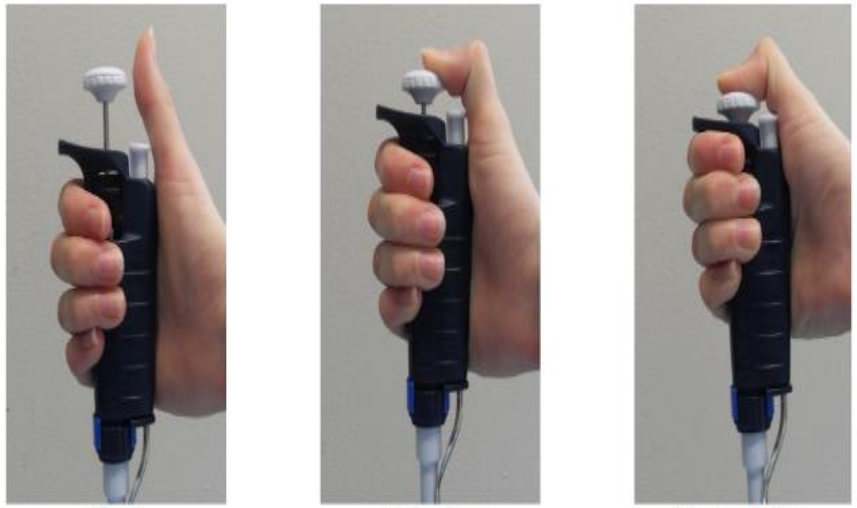
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Practical # 08

Liquid Handling: Proper Use of Micropipettes

Micropipette

- A commonly used in laboratories to aspire and dispense liquid, usually down to 0.1 uL.
- Micropipettes work by air displacement
- The operator depresses a plunger that moves
- an internal piston to one of two different positions. The first stop is used to fill the micropipette
- tip, and the second stop is used to dispense the contents of the tip. As the operator depresses the
- plunger to the first stop, an internal piston displaces a volume of air equal to the volume shown
- on the volume indicator dial. The second stop is used only to dispense the contents of the tip.



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Guidelines using a micropipette

- NEVER exceed the upper or lower limits of these pipettes. The limits are:

1. P10: 1.0 - 10.0µl
2. P20: 2.0 - 20.0µl
3. P200: 20 - 200µl
4. P1000: 200 - 1000µ





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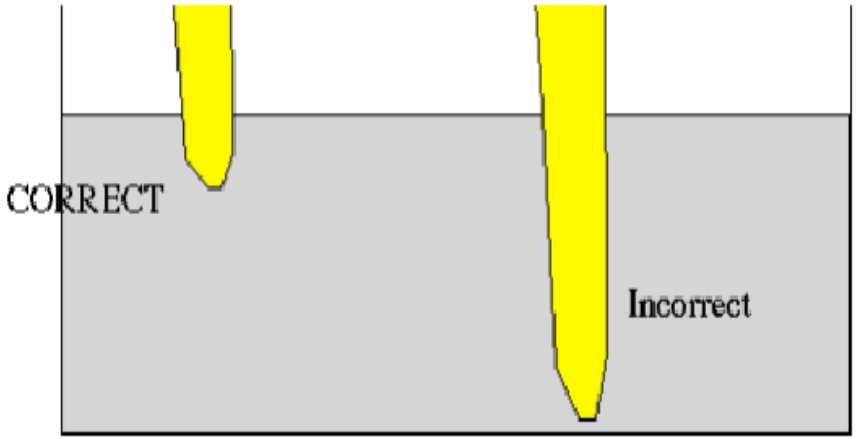
Steps of using micropipettes

- Always select the SMALLEST size pipet that will handle the volume you wish to move to achieve the greatest accuracy. Accuracy decreases as you use unnecessarily large pipets for small volumes.
- Set the desired volume by turning the centrally located rings clockwise to increase volume or counterclockwise to decrease volume.
- **P10**: Maximum volume 10µl. Accurate between 1µl and 10µl. Numbers on the micropipette (typically black-black-red) are read as XX.Xµl. The change in color indicates the position of the decimal point.
- **P20:** Maximum volume 20µl. Accurate between 2µl and 20µl. Numbers on the micropipette (typically black-black-red) are read as XX.Xµl. The change in color indicates the position of the decimal point.
- **P200**: Maximum volume 200µl. Accurate between 20µl and 200µl. Numbers on the micropipette (one color) are read as XXXµl.
- **P1000:** Maximum volume 1000µl (= 1 ml). Accurate between 200µl and 1000µl. Numbers on the micropipette (typically red-black-black) are read X.XX ml. Note that this micropipette reads milliliters while the other two read microliters.
- Load a sterile tip. Use blue tips for P1000 pipettes and clear tips for P200 and all smaller sizes. Use filter tips when performing PCR or working with RNA.
- Close the tip box to maintain sterility.

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Load of samples

- The plunger will stop at two different positions when it is depressed. Push the plunger down slowly to the point of first resistance: this is the load volume.
- Because this first stopping point is dependent on the volume that is being transferred, the distance you have to push the plunger to reach the point of initial resistance will change depending on the volume being pipetted.
- While holding the plunger at the load volume set point, put the tip into the solution so that it is immersed just enough to cover the end (3-4 mm), not as deep as possible.
- Slowly release the plunger to draw up the liquid making sure to keep the tip immersed. NOTE: If the solution you are pipetting is viscous, allow the pipet tip to fill to final volume before removing it from solution to avoid the presence of bubbles in the plastic tip, which will result in an inaccurate volume.
- Visually inspect the load to make sure it is correct - there should be no air space in the distal end tip



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Deliver the sample

- The second stopping point can be found when the plunger is depressed beyond the initial resistance until it is in contact with the body of the pipette.
- This second stopping point is used for the complete discharging of solutions from the plastic tip. You should not reach this second stop when drawing liquid into the pipette, only when expelling the last drop.

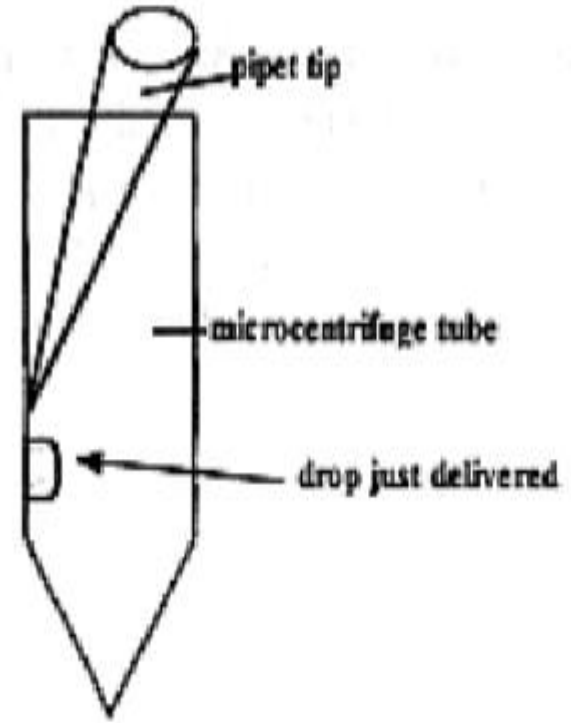
To deliver the volume,

- Place the tip into the receiving vessel.
- Depress the plunger to the point of initial resistance
- Wait 1 second
- Continue to press the plunger all the way to the bottom - this expels all the liquid.
- THEN, WITHOUT RELEASING THE PLUNGER, withdraw the tip.
- Discharge the tip. While holding the tip over an appropriate waste receptacle,
- press the discharge slider on the back of the grip.

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Small Volumes Technique

- With small volumes, especially the 1-10µl range, you must keep track of the droplets your pipet.
- Carefully expel the liquid droplet on the side wall of the tube so that you can see it, drawing the tip away/out carefully BEFORE releasing the plunger.
- If adding to a larger volume, flush the tip with the solvent liquid after expelling the droplet to make sure you get all the delivery liquid.
- With small volumes you'll usually need to centrifuge and then vortex the tube to get a good mixing of the reagents



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Calibration of micropipette

- Check the calibration of your micropipette by using the fact that 1 ml of deionized (or distilled) water has a mass of 1 g.
- Pipet a range of volumes spanning the pipette's useable range and weigh them on a top loading balance having at least 3 decimal place accuracy.
- Pipets having greater than 5% error should be recalibrated.



Practical # 09

Hematoxylin and Eosin Staining (H & E Staining)

What is H & E Staining?

- Most cells are colorless and transparent, and therefore histological (tissues) sections have to be stained in some way to make the cells visible.
- H & E Staining is a popular staining method used in histology and medical diagnosis
- For example, pathologist looks for cancer biopsy

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H & E stains

- H & E requires two histological stains

i. Hematoxylin

- Basic dye
- used in combination with aluminum (hemalum)
- stains acidic components i.e. nuclei which gives blue color

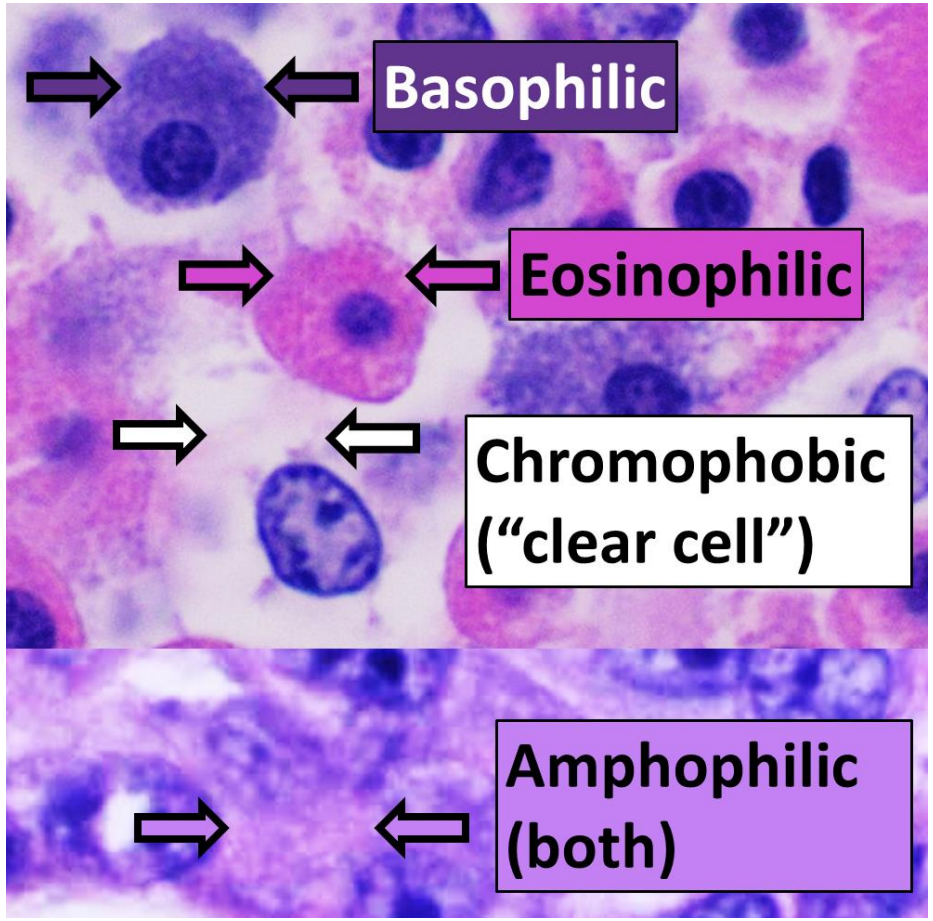
ii. Eosin

- Acidic dye
- stains basic components i.e. cytoplasm which gives red or pink color

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Hematoxylin and Eosin dyes



Hematoxylin and Eosin staining

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Solutions and reagents

1. Acid Alcohol Solution (1%)

- Hydrochloric acid, 1 ml
- 70% ethanol, 50 ml
- Mix well.

2. Ammonia Water Solution (0.2%)

- Ammonium hydroxide (concentrated), 2 ml
- Distilled water, 1000 ml
- Mix well.

3. **Lithium Carbonate Solution (Saturated)**

- Lithium carbonate 1.54 g
- Distilled water 100 ml

Mix well

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Eosin-Phloxine B Solution

Prepare the stock solutions first, and then create the working solution as needed.

1. Eosin Stock Solution

- Eosin Y, 1 g
- Distilled water, 100 ml
- Mix to dissolve.

2. **Phloxine Stock Solution**

- Phloxine B, 1 g
- Distilled water, 100 ml
- Mix to dissolve.

3. Eosin-Phloxine B Working Solution

Eosin stock solution, 100 ml

Phloxine stock solution, 10 ml

Ethanol (95%), 780 ml

Glacial acetic acid, 4 ml

Mix well.

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Hematoxylin Solution

- Potassium or ammonium (alum), 100 g Distilled water, 1000 ml

1. Heat to dissolve. Add 50 ml of 10% alcoholic hematoxylin solution and heat to boil for 1 minute.
2. Remove from heat and slowly add 2.5 g of mercuric oxide (red).
3. Heat to the solution and until it becomes dark purple color.
4. Cool the solution in cold water bath and add 20 ml of glacial acetic acid (concentrated).
5. Filter

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Staining procedure

1. Deparaffinization

- Deparaffinization sections, 2 changes of xylene, 10 minutes each.

2. Rehydration

- Re-hydrate in 2 changes of absolute alcohol, 5 minutes each, 95% alcohol for 2 minutes and 70% alcohol for 2 minutes and wash briefly in distilled water.

3. Hematoxylin staining

- Stain in Harris hematoxylin solution for 8 minutes and wash in running tap water for 5 minutes.

4. Differentiation

- Differentiate in 1% acid alcohol for 30 seconds and wash running tap water for 1 minute.

5. Bluing

- Bluing in 0.2% ammonia water or saturated lithium carbonate solution for 30 seconds to 1 minute and wash in running tap water for 5 minutes

6. Destaining

- Rinse in 95% alcohol, 10 dips.

7. Counterstain with Eosin

- Counterstain in eosin-phloxine solution for 30 seconds to 1 minute

8. Dehydration

- Dehydrate through 95% alcohol, 2 changes of absolute alcohol, 5 minutes each.

9. Clearing

- Clear in 2 changes of xylene, 5 minutes each.

10. Mounting

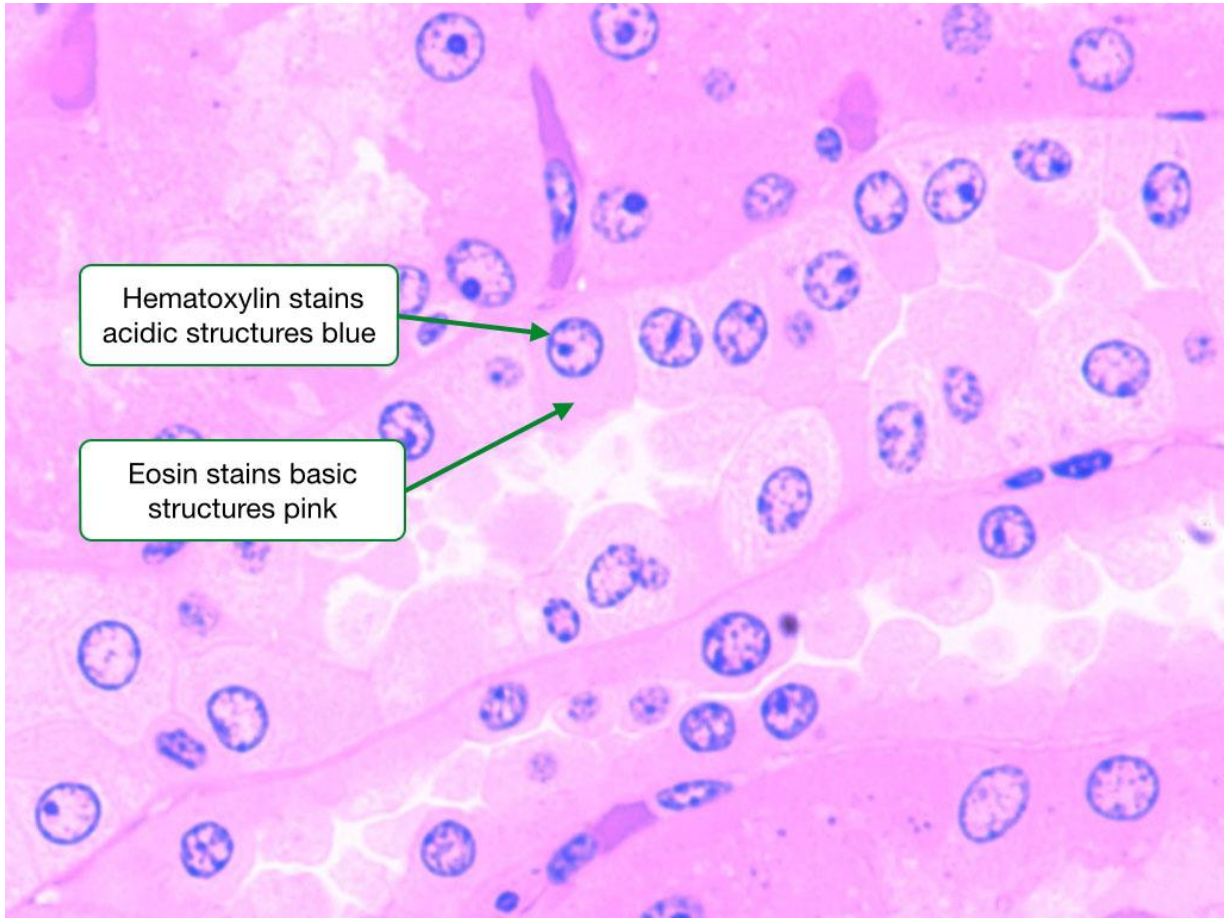
- Mount with xylene based mounting medium.

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Deparaffinistaion and rehydration	Xylene I	- 5 min
	Xylene II	- 5 min
	90% Alcohol	- 5 min
	70% Alcohol	- 5 min
	Water wash	- 10 min
Nuclear staining	Harri's hematoxylin	- 8 min
	Water wash	- 2 min
Differentiation	Differentiation in 1% Acid alcohol	- 1 dip
	Water wash	- 10 min
Bluing	1% Lithium carbonate	- 1 min
	Water wash	- 10 min
Cytoplasmic staining	1% Eosin	- 1 min
Dehydration	90% Alcohol	- 30 sec
	70% Alcohol	- 30 sec
	Xylene I	- 5 min
	Xylene II	- 5 min
	Approximate time required	- 70-75 min

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Results



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Practical # 11

Handling of Centrifuge Machines

What is centrifuge machine?

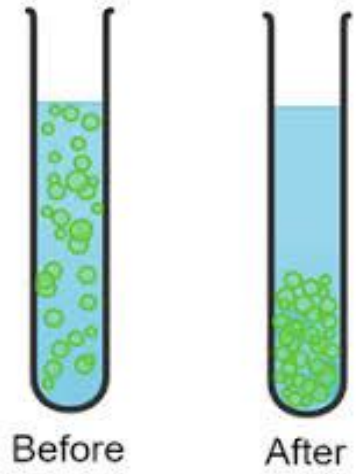
- A ubiquitous instrument in biomedical laboratories
- Applications include

i. The separation of serum or plasma from red blood cells,

ii. The separation of precipitated solids from the liquid phase of a mixture, or

iii. The separation of liquids of varying density

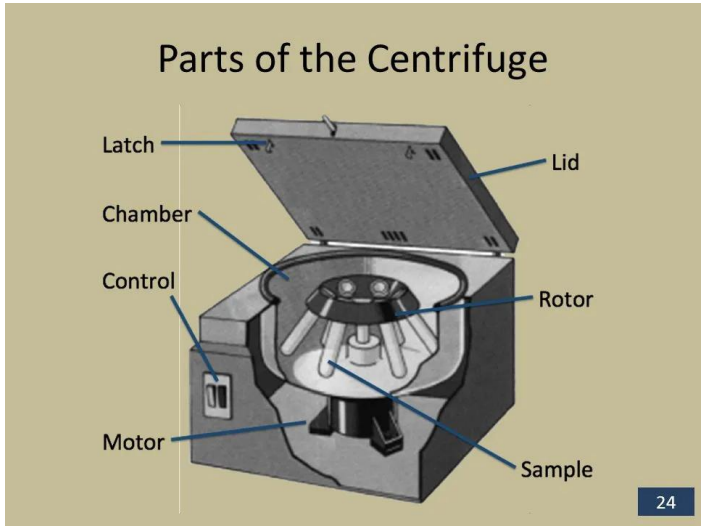




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Principles of centrifugation

- Particles suspended in a fluid move, under the influence of gravity, towards the bottom of a vessel at a rate that depends, in general, on their size and density.
- It utilizes centrifugal forces, greater than the force of gravity, to speed up the sedimentation rate of particles.
- This works by spinning the vessel containing the fluid and particles about an axis of rotation so that the particles experience a force acting away from the axis.
- The force is measured in multiples of the Earth‘s gravitational force and is known as t h e relative centrifugal field (RCF) or, more commonly, the x g‘ force.



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RPM vs RCF

RPM

- Revolutions Per Minute (RPM) is simply a measurement of how fast the centrifuge rotor does a full rotation in one minute.
- Basically, it is telling us how fast the rotor is spinning.

RCF

- Relative Centrifugal Force (RCF) or g-force is a measurement of the gravitational force that a sample is subject to.
- The force is generated from the spinning of the rotor which, in turn, exerts this force outward on the centrifuge tube

To convert RPM to RCF:

$$RCF = 11.18 \times r \times \left(\frac{RPM}{1{,}000} \right)^2$$

* *r = radius in cm*

Types of centrifuge machine



Low speed centrifuge

- Rotor speed less than 10,000 rpm
- Rotor requires no vacuum
- Used to separate serum or plasma from red blood cells, and to harvest and purify chemical precipitates, intact cells,



High Speed centrifuge

- Rotor speeds up to 21,000 rpm. RCFs of 120,000 x g are possible.
- Require vacuum and refrigeration facility
- Separation of a number of cell constituents and in the
- isolation and purification of viruses



Ultra high speed centrifuge

- 30,000 rpm and RCFs of over 600,000 x g
- solation and purification of membrane components such as endoplasmic reticulum

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Centrifuge components

1. Rotor

- most centrifuges allows the drive system to accept rot
- Three designs

i. Horizontal Rotor

ii. fixed angle rotor

iii. Vertical Rotor



Horizontal Rotor



Fixed angle rotor

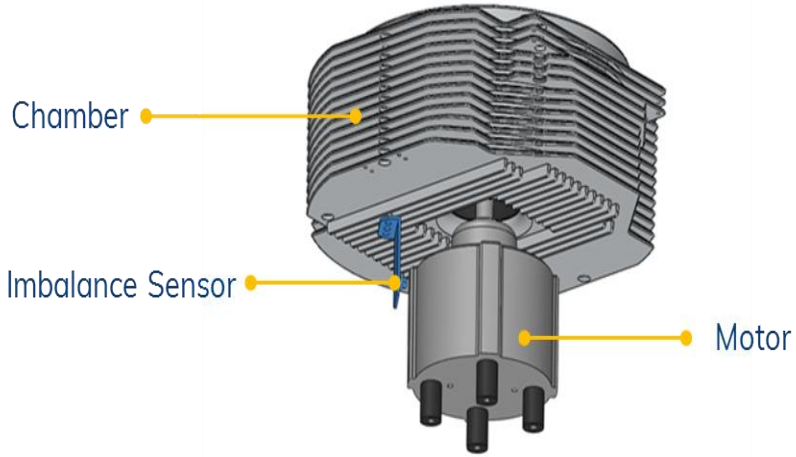


Vertical rotor

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2. Motor

- In general, centrifuge motors are high-torque, series-wound DC motors, the rotation of which increases as the voltage is increased
- Electrical contact to the commutator is provided by graphite brushes, which gradually wear down as they press against the commutator turning at high speed, and thus should be replaced at specified intervals.
- The speed of the centrifuge is controlled by a potentiometer that raises and lowers the voltage supplied to the motor.



3. Imbalance detector

Some instruments have an internal imbalance detector that monitors the rotor during operation, causing automatic shutdown if rotor loads are severely out of balance.

4. Tachometer

A tachometer indicates the speed in rpm. Most modem centrifuges use electronic tachometers, in which a magnet rotates around a coil to produce a current that can be measured.

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5. Safety lid:

Modem centrifuges must have a door-locking mechanism to prevent the lid from being opened while the instrument is running

6. Refrigerator

A centrifuge generates heat as it rotates and if samples are temperature labile then a refrigerated centrifuge should be used. Some centrifuges enable the rotor and chamber to be precooled before a run.

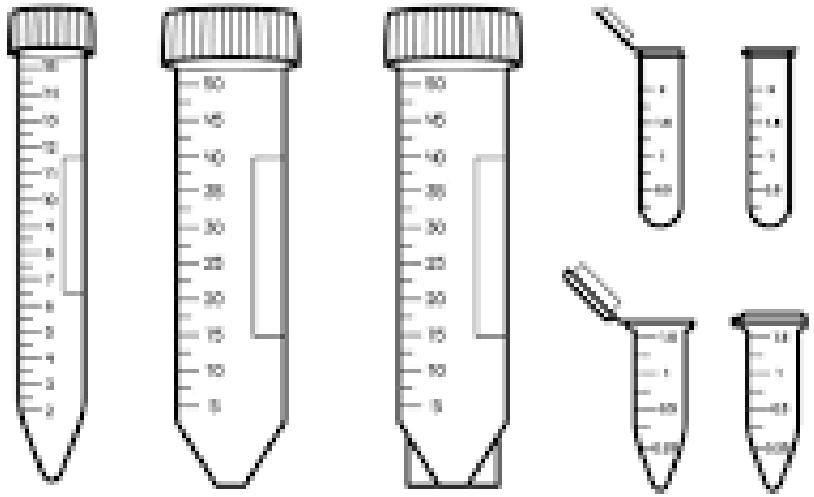


7. Braking system

Braking devices are incorporated to provide rapid rotor deceleration. Modern instruments have an electrical braking system that functions by reversing the polarity of the electrical current to the motor.

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Centrifuge tubes



Plastic Laboratory Centrifuge Tubes

Preventive measures

- Brushes should be removed regularly and checked for wear
- The rotor, buckets and shields or carriers should be examined for signs of mechanical stress (e.g. cracks, corrosion)
- Regularly lubricate the contact areas between the centrifuge buckets and the pins.
- Regularly check the condition of the O-ring on the tie-down nut on top of the rotor, and replace it if worn or damaged.
- Always follow a manufacturer's specific instructions.

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